Thông tin chung (General Instructions)

PRACTICAL EXAMINATION
Exam Duration: 85 minutes

Instructions

Read these instructions carefully before starting the exam:

- You can use the materials on your computer, notebooks, and textbooks.
- You are NOT allowed to use electronic devices and other methods to share data with others.
- Do not arbitrarily disconnect the FPTU_EXAMONLINE network while doing the test. If any network problems occur, immediately notify the exam supervisor for assistance.

Besides the conditions above, students must meet the following requirements:

- Use Visual Studio Code (VS Code) tool to do the exam.
- Using NodeJS platform version 22.11.0.
- Use the available browser software on your computer.

This part is very important — read it carefully and follow the instructions:

- You are provided with one file containing data named: database.json in the given directory.
- You must use the solution provided in advance via email and install the additional library included in the accompanying request.
- You are not allowed to download any additional libraries while taking the test.
- Violating one of the above, your exam is considered invalid.

Requirements

- Copy the file "database.json" from the given folder into your exam folder.
- Open two Terminal Command windows and run the commands below:

Terminal 1 — Start JSON Server (Backend API):

```
npx json-server --watch database.json --port 9999
```

Terminal 2 — Start React App:

```
npm start
```

> **Note:** 0 will be given to the work that:
>
> - Does not run JSON Server at http://localhost:9999
> - Does not configure the React App root at http://localhost:3000
> - Has incorrect component routing URLs

Question 1 — Class List Page (5 points)

Objective: Create a React page that displays the list of classes with filtering and navigation features.

Component routing at:

```
http://localhost:3000/classes
```

(0 points if incorrect URL)

Fetch data from:

Method	Endpoint
GET	http://localhost:9999/classes
GET	http://localhost:9999/subjects

Display Requirements:

- Show the classes in a table with the following columns: | No | Class Name | Subject Name | Lecturer | Enrolled Students | Status |
- Enrolled Students: count only students with enrollment status = "enrolled" in the enrollments array.
- Status badge: OPEN = green background, CLOSED = red background.
- Subject Name: look up from subjects list using subjectId - no hard-coded values.

Subject Filter (Dropdown):

- Extract unique subjects from subjects list - no hard-coded values.
- Options: All Subjects + list of subject names. Filter classes by selected subject.

Status Filter (Radio buttons):

- Options: All / OPEN / CLOSED. Filter classes based on selected status.

URL Query Parameter:

- Support: http://localhost:3000/classes?status=OPEN
- Support: http://localhost:3000/classes?status=CLOSED
- On page load, read the status param and pre-select the radio filter.

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Navigation Button:

- Each class row must have a button: "View Attendance Summary"
- Redirect to: http://localhost:3000/classes/:id/attendance-summary

Figure Demo:

[Hình ảnh: Rendered React UI. Top header bar (dark orange/brown gradient) with brand text "Attendance System" on the left and a "Classes" tab button. Page title "Class List" with subtitle "Manage and view attendance by class". A white filter card containing: label "Subject:" with a dropdown showing "All Subjects"; label "Status:" with three radio buttons "All" (selected/filled), "OPEN", "CLOSED". Below: text "Showing **4** of 4 classes". A table with a dark-brown header row: NO | CLASS NAME | SUBJECT | LECTURER | ENROLLED | STATUS | ACTION. Data rows: (1) 1 | **SE1701** | Project Management | Dr. Nguyen Minh Tuan | 3 | green "OPEN" badge | orange "View Summary" button; (2) 2 | **SE1702** | Project Management | MSc. Tran Thi Lan | 3 | green "OPEN" badge | "View Summary"; (3) 3 | **SE1703** | Software Development Project | Dr. Le Quoc Hung | 3 | green "OPEN" badge | "View Summary"; (4) 4 | **SE1704** | Software Testing | MSc. Pham Van Duc | 2 | red "CLOSED" badge | "View Summary".]

Figure 1: Class List — all 4 classes with subject names, enrolled count and colored status badges (OPEN=green, CLOSED=red)

[Hình ảnh: Same UI layout as Figure 1. The "Subject:" dropdown now shows "Project Management"; "Status:" radio "All" selected. Text: "Showing **2** of 4 classes for **Project Management**". Table shows two rows: (1) 1 | **SE1701** | Project Management | Dr. Nguyen Minh Tuan | 3 | green "OPEN" | "View Summary"; (2) 2 | **SE1702** | Project Management | MSc. Tran Thi Lan | 3 | green "OPEN" | "View Summary".]

Figure 2: Filter by subject = Project Management — shows SE1701 and SE1702 only

[Hình ảnh: Same UI layout. "Subject:" dropdown = "All Subjects"; "Status:" radio "OPEN" selected (filled). Text: "Showing **3** of 4 classes with status **OPEN**". Table shows three rows, all with green "OPEN" badges: (1) 1 | **SE1701** | Project Management | Dr. Nguyen Minh Tuan | 3; (2) 2 | **SE1702** | Project Management | MSc. Tran Thi Lan | 3; (3) 3 | **SE1703** | Software Development Project | Dr. Le Quoc Hung | 3; each row has a "View Summary" button.]

Figure 3: Filter by status = OPEN (radio selected) — shows SE1701, SE1702, SE1703

[Hình ảnh: Same UI layout. "Subject:" dropdown = "All Subjects"; "Status:" radio "CLOSED" selected (filled). Text: "Showing **1** of 4 classes with status **CLOSED**". Table shows one row: 1 | **SE1704** | Software Testing | MSc. Pham Van Duc | 2 | red "CLOSED" badge | "View Summary" button.]

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Figure 4: Filter by status = CLOSED — shows SE1704 only with red badge

[Hình ảnh: A light-blue info box with the orange heading "URL Query Parameter Support:". Inside, an orange square button with a left/back arrow (←) followed by the orange link text "http://localhost:3000/classes?status=CLOSED", illustrating that loading this URL pre-selects the CLOSED radio filter.]

Figure 5: URL query param ?status=CLOSED pre-selects the CLOSED radio on page load

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Question 2 — Attendance Summary Page (5 points)

Objective: Create a React page that displays attendance statistics for all students in a specific class.

Component routing at:

```
http://localhost:3000/classes/:id/attendance-summary
```

(0 points if incorrect URL)

Fetch data from:

Method	Endpoint
GET	http://localhost:9999/classes/:id
GET	http://localhost:9999/subjects/:id
GET	http://localhost:9999/attendanceSummaries?classId=xxx
GET	http://localhost:9999/students

Display Requirements:

- Show class name and subject name at the top of the page.
- Show attendance statistics in a table with the following columns: | No | Student Code | Full Name | Total Sessions | Present | Absent | Attendance Rate | Status |
- Attendance Rate: display as percentage rounded to 1 decimal (e.g. 90.0%).
- Status badge: attendanceRate < 80% → "Warning" in red; otherwise "OK" in green.
- Highlight the entire row in light red (#FFF5F5) if attendanceRate < 80%.

Summary Stats (above the table):

- Total number of students enrolled in the class.
- Number of students with warning (attendanceRate < 80%).
- Average attendance rate of the class (rounded to 1 decimal %).

Sort Feature:

- Click on "Att. Rate" column header to toggle sort: no sort → ascending (↑) → descending (↓) → no sort.
- Show sort indicator: ↑ for ascending, ↓ for descending, ↕ for no sort.
- Sorted column header text changes color to highlight active sort.

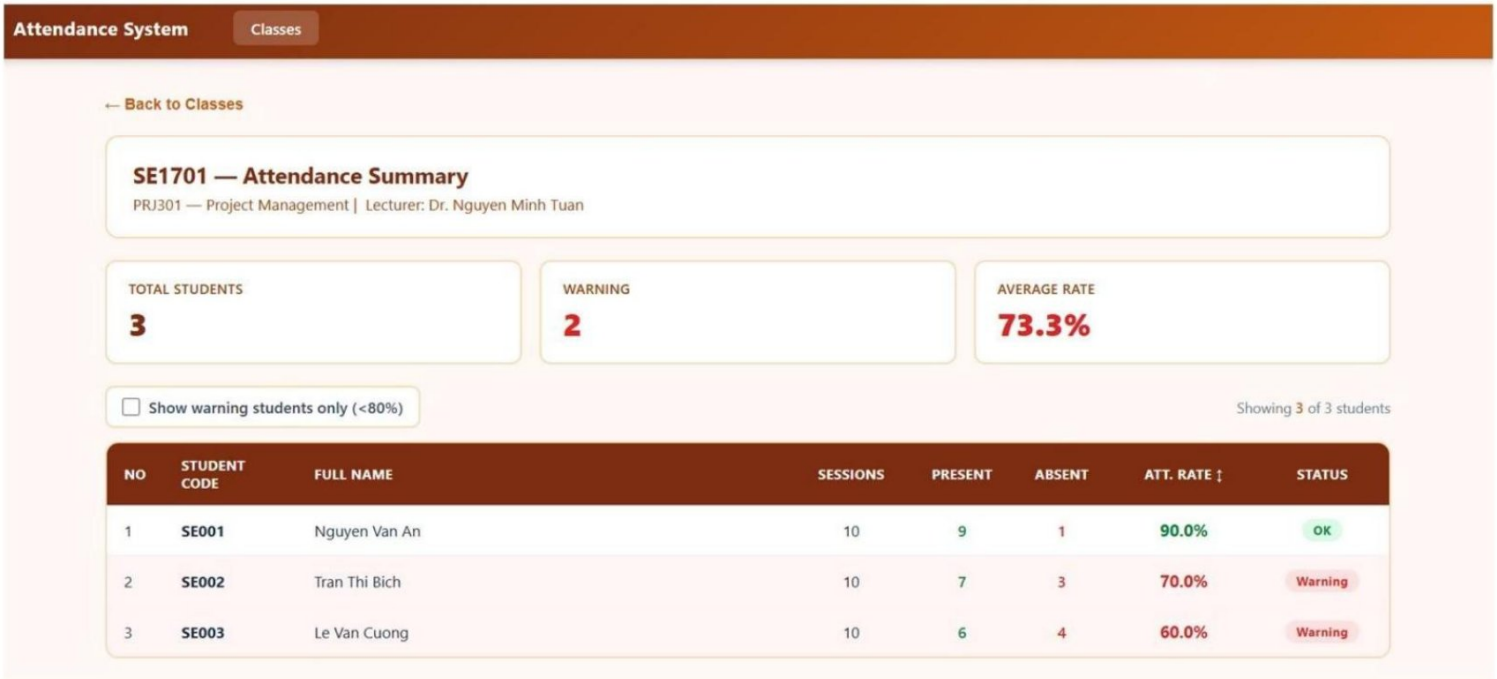
Warning Filter (Checkbox):

- Label: "Show warning students only (<80%)"
- When checked: hide students with attendanceRate ≥ 80%, showing only warning students.
- Result count updates to reflect filtered rows.

Navigation:

- "← Back to Classes" button → navigate back to http://localhost:3000/classes

Figure Demo:



[Hình ảnh: Rendered React UI. Header bar "Attendance System" + "Classes" tab. Link "← Back to Classes". A card with title "SE1701 — Attendance Summary" and subtitle "PRJ301 — Project Management | Lecturer: Dr. Nguyen Minh Tuan". Three summary stat cards in a row: "TOTAL STUDENTS" = **3**; "WARNING" = **2** (red number); "AVERAGE RATE" = **73.3%** (red number). Below: an unchecked checkbox labeled "Show warning students only (<80%)" on the left, and right-aligned text "Showing **3** of 3 students". A table with dark-brown header: NO | STUDENT CODE | FULL NAME | SESSIONS | PRESENT | ABSENT | ATT. RATE ↕ | STATUS. Rows: (1) 1 | **SE001** | Nguyen Van An | 10 | 9 (green) | 1 (red) | 90.0% (green) | green "OK" badge; (2) 2 | **SE002** | Tran Thi Bich | 10 | 7 (green) | 3 (red) | 70.0% (red) | red "Warning" badge (row highlighted light red); (3) 3 | **SE003** | Le Van Cuong | 10 | 6 (green) | 4 (red) | 60.0% (red) | red "Warning" badge (row highlighted light red).]

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Figure 6: SE1701 summary — stats bar (Total:3, Warning:2, Avg:73.3%), default order, Warning badge for SE002(70%) and SE003(60%), warning rows highlighted red

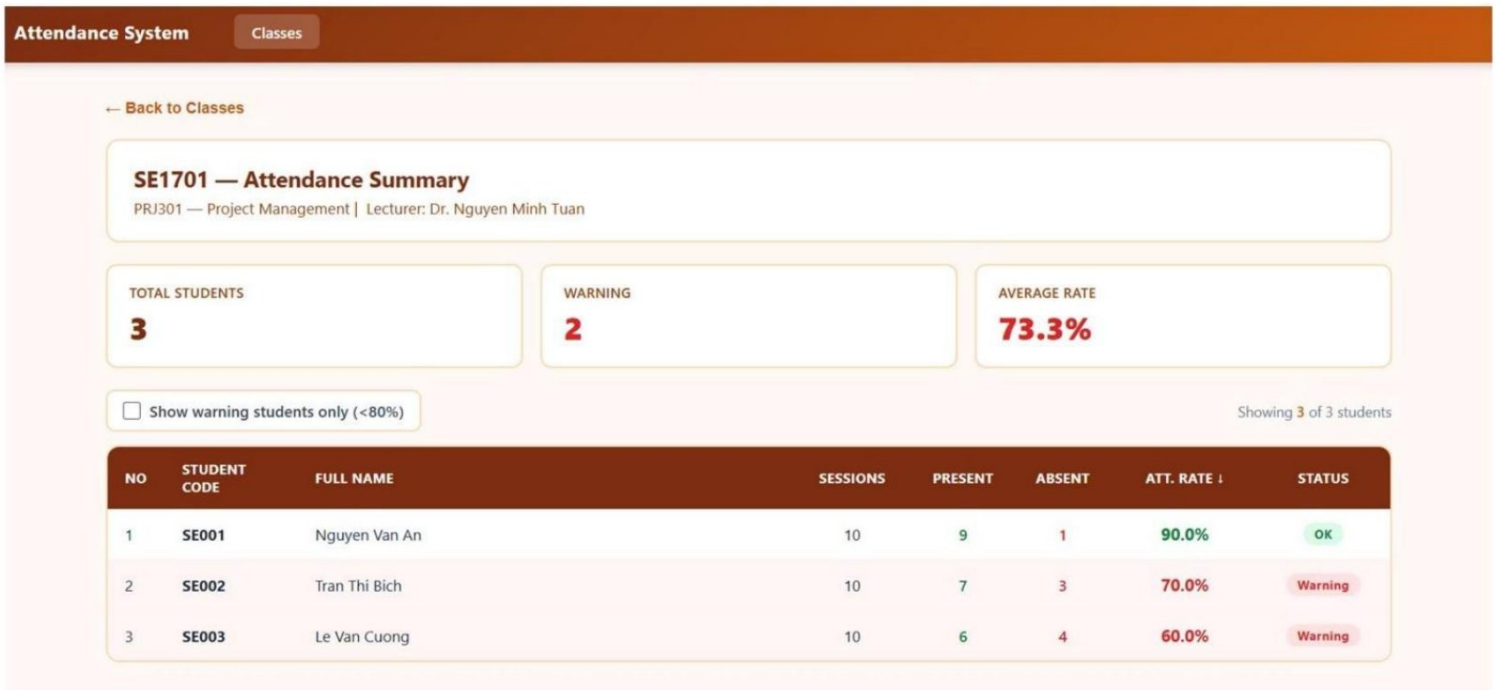




[Hình ảnh: Same page as Figure 6 (same header card and stat cards Total 3 / Warning 2 / Average 73.3%). The "ATT. RATE" column header shows an up arrow (↑) and is highlighted. Rows sorted ascending (lowest rate first): (1) 1 | **SE003** | Le Van Cuong | 10 | 6 | 4 | 60.0% (red) | "Warning"; (2) 2 | **SE002** | Tran Thi Bich | 10 | 7 | 3 | 70.0% (red) | "Warning"; (3) 3 | **SE001** | Nguyen Van An | 10 | 9 | 1 | 90.0% (green) | "OK".]

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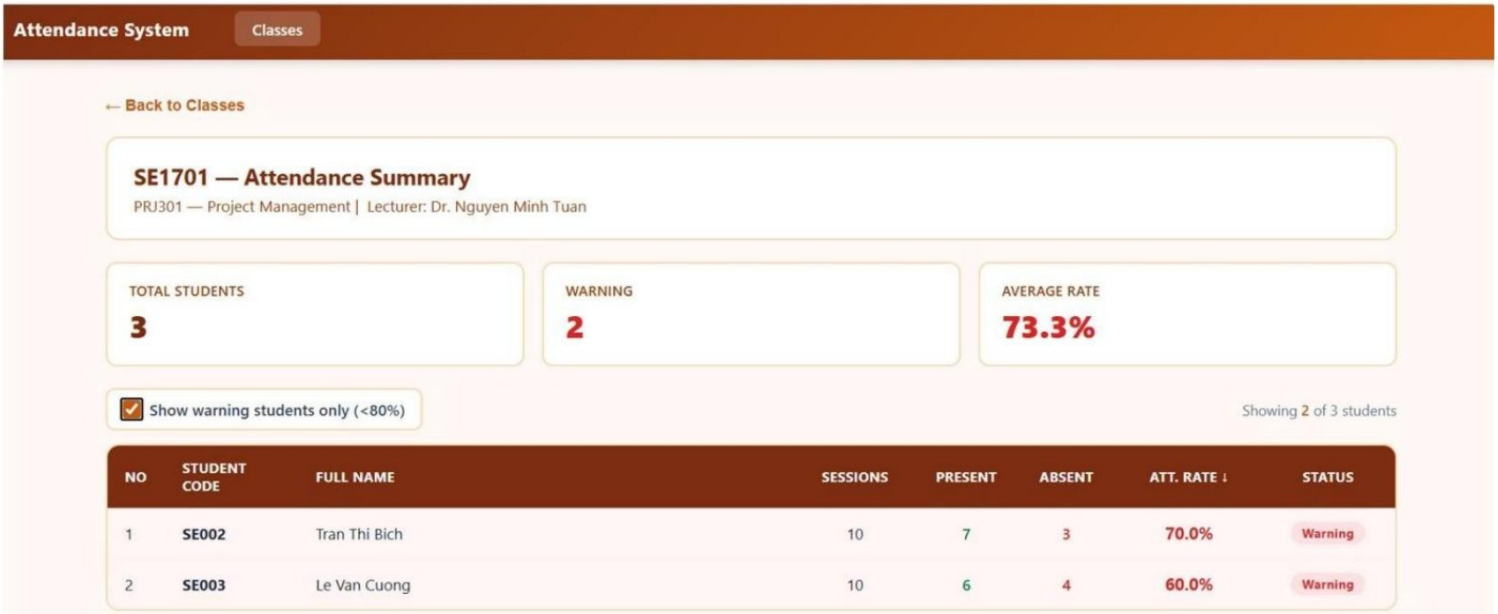
Figure 7: Sort ascending ↑ — lowest rate first: SE003(60%) → SE002(70%) → SE001(90%), column header highlighted



[Hình ảnh: Same page as Figure 6/7. The "ATT. RATE" column header shows a down arrow (↓). Rows sorted descending (highest rate first): (1) 1 | **SE001** | Nguyen Van An | 10 | 9 | 1 | 90.0% (green) | "OK"; (2) 2 | **SE002** | Tran Thi Bich | 10 | 7 | 3 | 70.0% (red) | "Warning"; (3) 3 | **SE003** | Le Van Cuong | 10 | 6 | 4 | 60.0% (red) | "Warning".]

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Figure 8: Sort descending ↓ — highest rate first: SE001(90%) → SE002(70%) → SE003(60%)

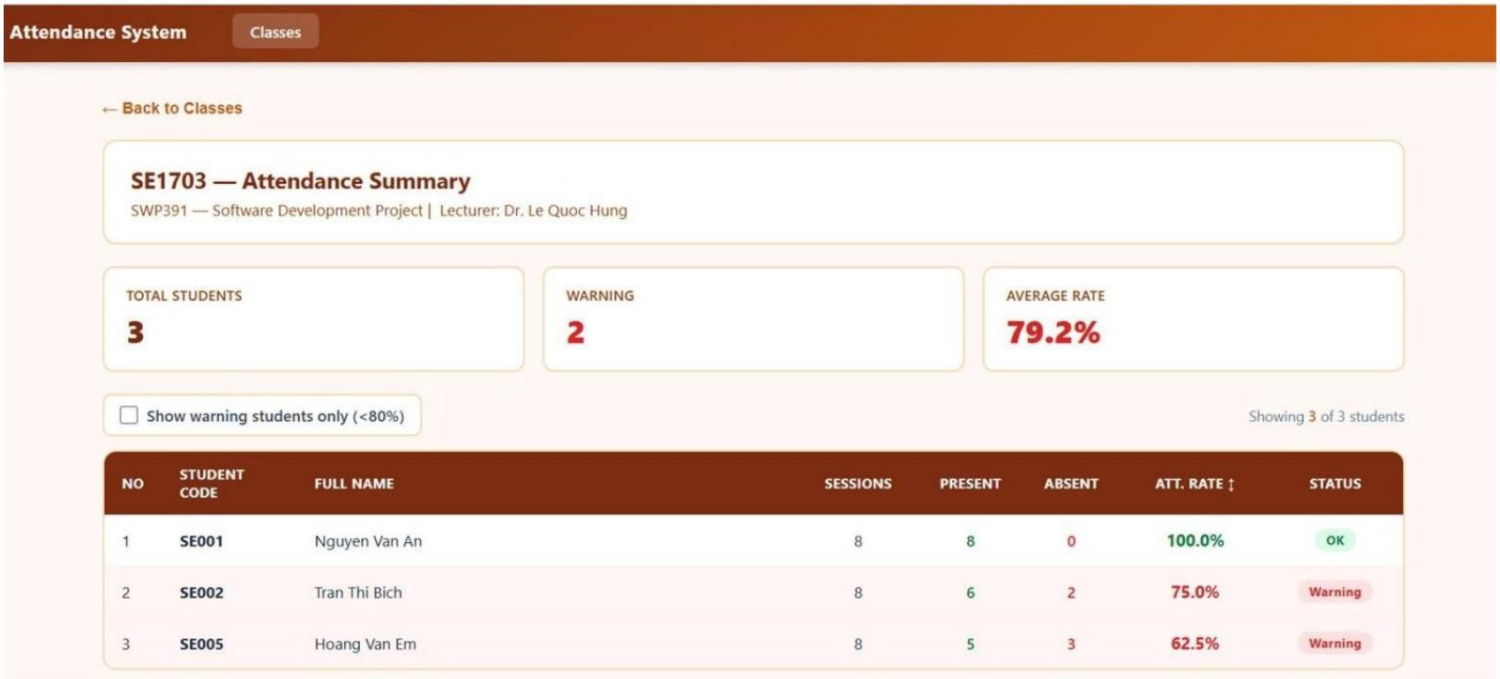


[Hình ảnh: Same page as Figure 6. The checkbox "Show warning students only (<80%)" is now CHECKED (filled orange). Right-aligned text reads "Showing **2** of 3 students". "ATT. RATE" header shows down arrow (↓). The table shows only the two warning rows (OK row hidden): (1) 1 | **SE002** | Tran Thi Bich | 10 | 7 | 3 | 70.0% (red) |

"Warning"; (2) 2 | **SE003** | Le Van Cuong | 10 | 6 | 4 | 60.0% (red) | "Warning".]

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Figure 9: Warning filter ON — checkbox checked, only SE002 and SE003 shown, result count shows "2 of 3 students"



[Hình ảnh: Same page layout but for class SE1703. Card title "SE1703 — Attendance Summary", subtitle "SWP391 — Software Development Project | Lecturer: Dr. Le Quoc Hung". Stat cards: "TOTAL STUDENTS" = **3**; "WARNING" = **2** (red); "AVERAGE RATE" = **79.2%** (red). Checkbox unchecked; right text "Showing **3** of 3 students". "ATT. RATE" header shows up-down arrow (↕, no sort). Rows: (1) 1 | **SE001** | Nguyen Van An | 8 | 8 (green) | 0 (red) | 100.0% (green) | green "OK" badge; (2) 2 | **SE002** | Tran Thi Bich | 8 | 6 (green) | 2 (red) | 75.0% (red) | red "Warning" badge (row light red); (3) 3 | **SE005** | Hoang Van Em | 8 | 5 (green) | 3 (red) | 62.5% (red) | red "Warning" badge (row light red).]

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Figure 10: SE1703 summary — different class: SE001(100%/OK), SE002(75%/Warning), SE005(62.5%/Warning), Average=79.2%

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